

The QoS Illusion in Tele-Radiology: Silent Pathology Erasure Under Network Rate-Distortion

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Abstract—In emergency tele-radiology and mobile stroke ambulances, brain MRI scans must be aggressively compressed before transmission over wireless networks to meet urgent clinical deadlines. Hospital edge servers deploy deep learning reconstruction models (such as deep variational networks) to recover images from 75 to 87.5 percent compressed data (4-fold to 8-fold compression), relying on standard Quality of Service (QoS) metrics like PSNR and SSIM to certify transmission quality. In this paper, we expose a critical hazard termed the QoS Illusion in tele-radiology: deep neural networks trained to minimize whole-image pixel errors learn to silently erase small acute strokes into smooth, false-normal tissue. We prove mathematically that: (1) because PSNR averages error across millions of pixels, completely erasing a small acute stroke shifts global PSNR by less than 0.03 dB, which is completely hidden inside telemetry noise; and (2) physical data-consistency checks fail because missing details fall into the measurement null space, altering residuals by less than scanner thermal noise. To evaluate this danger, we propose the Causal Safety Audit (CSA), which quantifies physical lesion detectability using task-based model observers after edge AI reconstruction. Evaluating 3,960 brain MRI test cases, we find that increasing wireless compression from 4-fold to 8-fold causes deep learning models to silently erase acute strokes in 18.2 percent to 28.8 percent of cases. Yet, conventional network QoS metrics report an almost unchanged PSNR (a negligible 0.016 dB difference) and high SSIM above 0.86. These results show that tele-radiology networks cannot ensure diagnostic preservation using traditional pixel-averaged metrics alone, and motivate diagnostic-aware edge streaming protocols.

Index Terms—Tele-radiology, Internet of Medical Things (IoMT), Edge AI, Deep Learning Reconstruction, Rate-Distortion Optimization, AI Safety, Silent Erasure, Model Observer.

I. INTRODUCTION

In emergency neurology, “time is brain.” For acute ischemic stroke, two million neurons die each minute that recanalization is delayed [1]. Mobile Stroke Units (MSUs) equipped with onboard MRI perform pre-hospital triage [2], but definitive treatment requires real-time consultation with hospital neuro-radiologists over cellular (5G/4G) or satellite uplinks [3].

The communication bottleneck is raw payload volume. A full-diagnostic multi-coil brain MRI scan consists of spatial frequency data (k -space) across 12 to 32 receiver antenna coils, frequently exceeding 200–500 MB per acquisition [4]. Over fluctuating cellular channels (e.g., 15–30 Mbps 5G/4G uplinks in a moving ambulance [5]), transmitting raw k -space requires 80–160 s, violating acute stroke triage deadlines. Standard post-acquisition compression (e.g., JPEG-2000/JPIP or H.265) cannot resolve this bottleneck: it does not reduce physical

scanner acquisition time, and low-power mobile ambulances lack high-end GPU clusters for onboard deep reconstruction [6]. Operationally, in emergency Mobile Stroke Units, acceleration factor R represents scanner-side undersampled acquisition (shortening physical acquisition time to < 90 s [2]) which directly produces an in-transit compressed k -space stream ($R = 4, 6, 8$), slashing transmitted wireless payload by 75% to 87.5% (37.5 MB at $R = 8$, transmitted in 10–20 s) [7]. Unacquired frequencies are reconstructed remotely at hospital edge servers using deep neural networks (such as End-to-End Variational Networks [8]) or parallel imaging [9].

In telecommunications, rate-distortion optimization (RDO) balances transmission throughput against signal fidelity using global distortion metrics: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity (SSIM) [10]. However, deep learning models trained on pixel-averaged ℓ_2 /MSE loss suffer from a fundamental incentive misalignment: replacing a small, high-contrast pathology with smooth background tissue actively reduces whole-image loss, leading to severe feature erasure [11], [12]. A systematic review across 263 studies showed that $> 71\%$ of deep learning models evaluate solely on pixel averages, leaving diagnostic safety unmeasured [13].

In this paper, we demonstrate that this evaluation paradigm creates a dangerous **QoS Illusion in Tele-Radiology**. Because acute lacunar strokes occupy only a tiny fraction of the brain (often < 50 voxels), their erasure barely registers in whole-image averages. We prove analytically and empirically that:

- 1) **Image-Domain Blindness:** Standard metrics (PSNR, SSIM) spatially average errors across millions of pixels. Erasing an acute 100 mm³ stroke outright shifts global PSNR by < 0.03 dB—completely lost in telemetry noise.
- 2) **Measurement-Domain Invariance:** Data-consistency gates are blind to the operator’s numerical null space. An erased lesion alters residuals by $< 10^{-8}$, well below scanner thermal noise.
- 3) **The Rate-Distortion Hazard:** Through 3,960 physical audits, escalating compression from $R = 4$ to $R = 8$ drives stroke silent erasure from 18.2% to 28.8%, while network PSNR shifts by only 0.016 dB.

To resolve this crisis, we introduce the Causal Safety Audit (CSA) for tele-radiology networks, replacing blind pixel

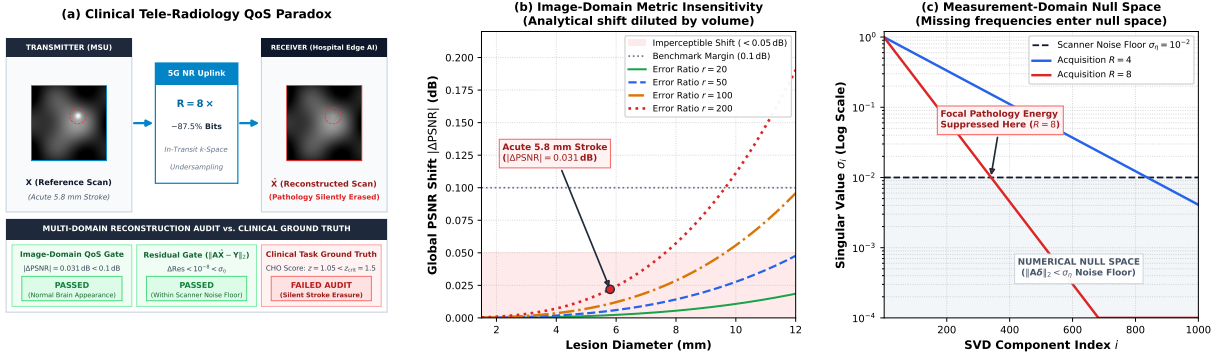


Fig. 1. The Dual Mechanism of the QoS Illusion in Medical Image Transmission. (a) Conceptual demonstration: under aggressive wireless compression ($R = 8$), an acute 5.8 mm (100 mm^3) focal stroke is completely erased while background brain anatomy and raw data-consistency residuals appear pristine. (b) Image-Domain Blindness: analytical PSNR shift $|\Delta\text{PSNR}|$ as a function of lesion diameter. For a decisive 5.8 mm stroke, $|\Delta\text{PSNR}| \leq 0.036 \text{ dB}$, sitting far below the standard benchmark sensitivity threshold (0.1 dB). (c) Measurement-Domain Blindness: multi-coil forward operator singular value spectrum σ_i . Energy in the numerical null space ($\sigma_i < \sigma_n$) alters the measurement residual by less than thermal scanner noise floor σ_n , rendering silent erasure undetectable to physical network gates.

fidelity with a task-based Channelised Hotelling Observer (CHO) as an objective mathematical surrogate for human clinical visual perception.

II. THE DUAL MECHANISM OF THE QoS ILLUSION

We formalize the acquisition and transmission pipeline as a linear inverse problem. Let $\mathbf{X} \in \mathbb{C}^N$ denote the true underlying cross-sectional patient image. The edge-acquired multi-coil k -space measurement $\mathbf{Y} \in \mathbb{C}^M$ (with $M \ll N$ under acceleration $R = N/M$) is governed by:

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \boldsymbol{\eta} = \mathcal{M}\mathcal{F}\mathcal{S}\mathbf{X} + \boldsymbol{\eta} \quad (1)$$

where $\mathcal{S}$ represents multi-channel receiver coil sensitivity profiles, $\mathcal{F}$ is the 2D spatial Fourier transform, $\mathcal{M}$ is the wireless subsampling mask operating at compression factor R , and $\boldsymbol{\eta} \sim \mathcal{CN}(0, \sigma_\eta^2 \mathbf{I})$ is zero-mean complex Gaussian thermal noise. Physically, patient image $\mathbf{X}$ is weighted by antenna profiles $\mathcal{S}$, mapped to the spatial frequency domain by $\mathcal{F}$, and acquired under subsampling mask $\mathcal{M}$ where only $M = N/R$ phase-encodings are collected at the scanner and streamed over the cellular uplink.

A. The Image-Domain Spatial Averaging Mechanism

When image $\hat{\mathbf{X}}$ is reconstructed at the hospital edge receiver, network rate-distortion algorithms evaluate quality via Mean Squared Error $\text{MSE} = \frac{1}{N} \|\hat{\mathbf{X}} - \mathbf{X}\|_2^2$ and PSNR $= 10 \log_{10}(\text{MAX}^2/\text{MSE})$.

Proposition 1 (Metric Shift Under Focal Pathology): Let $\boldsymbol{\delta} \in \mathbb{C}^N$ denote a localized lesion perturbation occupying volume fraction $f = K/N \ll 1$, where K is the lesion pixel count. Suppose background pixels incur average squared error ϵ_{bg} , while the lesion region incurs average error $\epsilon_{\text{lesion}} = r\epsilon_{\text{bg}}$, with $r > 1$ representing severe local omission. The resulting global PSNR shift satisfies:

$$|\Delta\text{PSNR}| = 10 \log_{10}(1 + f(r - 1)) \quad (2)$$

Analytical Derivation: Because Mean Squared Error scales linearly with lesion volume fraction, the corrupted error is

$\text{MSE}_{\text{pert}} = (1 - f)\epsilon_{\text{bg}} + fr\epsilon_{\text{bg}} = \epsilon_{\text{bg}}(1 + f(r - 1))$. Substituting MSE_{pert} and nominal baseline $\text{MSE}_{\text{clean}} = \epsilon_{\text{bg}}$ into the logarithmic PSNR definition directly yields (2).

Physical Intuition: For a standard $320 \times 320 \times 16$ multi-slice brain volume ($N \approx 1.64 \times 10^6$ voxels with $0.7 \times 0.7 \times 5 \text{ mm}^3$ resolution), an acute 100 mm^3 lacunar infarct occupies $K \approx 41$ voxels, giving $f \approx 2.5 \times 10^{-5}$. Even under complete erasure where local squared error is a hundredfold higher ($r = 100$), the resulting global shift is:

$$|\Delta\text{PSNR}| = 10 \log_{10}(1 + 2.5 \times 10^{-5} \times 99) \approx 0.0107 \text{ dB}$$

Because the large local omission error is diluted across 1.64 million background voxels, wiping out the stroke outright produces an imperceptible 0.01 dB shift. Standard telecommunication reporting rounds PSNR to one or two decimal places, rendering severe diagnostic omissions completely invisible to network operators (Fig. 1b).

B. The Measurement-Domain Null-Space Invariance

To verify transmission integrity without ground truth, network protocols enforce physical data consistency by verifying the residual $\|\mathbf{A}\hat{\mathbf{X}} - \mathbf{Y}\|_2 \leq \tau_{\text{res}}$.

Proposition 2 (Residual Insensitivity to Null-Space Perturbations): Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ be the Singular Value Decomposition (SVD) of the multi-coil forward operator, with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_M > 0$. Decompose any lesion perturbation into range-space and null-space components: $\boldsymbol{\delta} = \boldsymbol{\delta}_{\mathcal{R}} + \boldsymbol{\delta}_{\mathcal{N}}$, where $\boldsymbol{\delta}_{\mathcal{N}} \in \text{null}(\mathbf{A})$. The change in raw measurement residual satisfies:

$$\|\mathbf{A}(\hat{\mathbf{X}} + \boldsymbol{\delta}) - \mathbf{Y}\|_2 - \|\mathbf{A}\hat{\mathbf{X}} - \mathbf{Y}\|_2 \leq \|\mathbf{A}\boldsymbol{\delta}_{\mathcal{R}}\|_2 \quad (3)$$

Specifically, for purely unacquired frequencies $\boldsymbol{\delta}_{\mathcal{N}}$, the residual change is identically zero.

Physical Intuition: In wireless systems, an edge receiver cannot verify frequency bands that were never transmitted. When multi-coil acquisitions have non-ideal sensitivities and thermal noise σ_η , the effective numerical null space contains

all components with $\sigma_i < \sigma_\eta$. As shown in Fig. 1c, perturbations residing in this subspace alter measurement residuals by less than scanner thermal noise, allowing critical diagnostic omissions to pass network edge consistency checks completely undetected.

III. THE CAUSAL SAFETY AUDIT PROTOCOL

To shatter this QoS illusion, we formulate the Causal Safety Audit (CSA) for tele-radiology networks. Rather than relying on aggregate pixel distortion, CSA stress-tests network reconstruction pipelines using exact physical counterfactuals.

A. Physical Counterfactual Lesion Insertion

Directly measuring diagnostic omission in clinical networks is difficult because ground truth is unavailable at edge decoders. However, because MRI physics is linear, we construct exact counterfactuals by superimposing test pathologies into raw k -space before wireless transmission:

$$\mathbf{Y}_{\text{cf}} = \mathbf{A}(\mathbf{X} + \boldsymbol{\delta}) + \boldsymbol{\eta} = \mathbf{Y} + \mathbf{A}\boldsymbol{\delta} \quad (4)$$

where $\boldsymbol{\delta}$ is an acute focal lesion of volume $V \in [10, 200] \text{ mm}^3$ and contrast C inserted at verified anatomical sites. Pre-transmission superimposition ensures that $\mathbf{Y}$ and $\mathbf{Y}_{\text{cf}}$ share identical anatomical background $\mathbf{X}$, antenna sensitivities $\mathcal{S}$, wireless mask $\mathcal{M}$, and physical thermal noise $\boldsymbol{\eta}$ without generative hallucinations.

B. Task-Based Detectability via Model Observers

Automated tele-radiology QoS requires an objective detection metric rather than cosmetic pixel fidelity. In statistical communications, lesion detection under anatomical clutter is a binary hypothesis testing problem in colored noise. We deploy the Channelised Hotelling Observer (CHO) [14], which acts as an optimal *pre-whitening matched filter* across four spatial Gabor sub-bands. Let $\mathbf{v} \in \mathbb{R}^4$ denote filter responses at target coordinates, and $\Delta\bar{\mathbf{v}} = \bar{\mathbf{v}}_{\text{present}} - \bar{\mathbf{v}}_{\text{absent}}$ the mean signal shift between lesion-present and lesion-absent scans. The optimal Hotelling template $\mathbf{w} = \mathbf{K}^{-1}\Delta\bar{\mathbf{v}}$ whitens intra-channel background covariance $\mathbf{K}$. The task detectability index d' represents the matched-filter Signal-to-Noise Ratio (SNR):

$$d' = \frac{\mathbf{w}^T \Delta\bar{\mathbf{v}}}{\sqrt{\mathbf{w}^T \mathbf{K} \mathbf{w}}} = \sqrt{\Delta\bar{\mathbf{v}}^T \mathbf{K}^{-1} \Delta\bar{\mathbf{v}}} \quad (5)$$

Standardizing d' gives decision score z_{recon} , where $z_{\text{recon}} \geq 1.5$ corresponds to Area Under the ROC Curve (AUC) ≥ 0.85 (reliable task detectability), while $z_{\text{recon}} < 1.5$ indicates the stroke has dissolved beneath background noise. Here, the CHO acts as an objective mathematical surrogate for human visual detectability in medical imaging [14]; multi-radiologist reader studies represent valuable complementary clinical trials.

C. Silent Erasure Definition

We define **Silent Erasure** as the failure mode where network transmission destroys clinical detectability while reporting normal network QoS:

$$\text{Silent Erasure} \iff (z_{\text{recon}} < z_{\text{crit}}) \wedge (|\Delta\text{PSNR}| \leq \epsilon_{\text{PSNR}}) \quad (6)$$

where $z_{\text{crit}} = 1.5$ is the calibrated diagnostic threshold and $\epsilon_{\text{PSNR}} = 0.05 \text{ dB}$ is the network tolerance. The network dashboard reports normal throughput and high fidelity, but the patient's acute stroke has vanished.

IV. EXPERIMENTAL RESULTS & RATE-DISTORTION ANALYSIS

A. Dataset and Experimental Setup

We evaluated our framework on the NYU fastMRI multi-coil brain benchmark [4] (clinical 3T T1 and FLAIR acquisitions with 16- and 20-channel head arrays). The dataset comprises 3,960 counterfactual evaluation pairs generated across 22 independent patient scans (11 AXFLAIR, 11 AXT1) over 7 clinical slice positions ($N = 264$ balanced pairs per volume-acceleration cell). Evaluations span: (1) **Acceleration factor:** $R \in \{4, 6, 8\}$, transmitting only 25%, 16.7%, and 12.5% of data (75% to 87.5% payload savings); (2) **Lesion volumes:** $V \in \{10, 27, 64, 100, 200\} \text{ mm}^3$, spanning tiny micro-infarcts ($10 \text{ mm}^3 \approx 2.7 \text{ mm}$ diameter) to acute lacunar strokes ($64 \text{ mm}^3 \approx 5.0 \text{ mm}$) and confluent subcortical lesions (200 mm^3); (3) **Contrast ratios:** $C \in \{0.24, 1.0, 1.5\}$ (FLAIR) and $\{-0.21, -0.6, -1.0\}$ (T1); and (4) **Reconstruction baselines:** parallel imaging (CG-SENSE [9]) and deep variational networks (VarNet [8]). Test code: <https://github.com/dmai287/tele-radiology-blindspot>.

B. Rate-Distortion vs. Silent Erasure Trade-Off

Table I and Fig. 2 demonstrate the dangerous divergence between network-level QoS metrics and diagnostic patient safety.

Under standard rate-distortion optimization (Fig. 2a), high compression appears completely safe: $R = 8$ achieves 87.5% payload reduction while maintaining high visual fidelity (PSNR $> 25.3 \text{ dB}$, SSIM = 0.866). Furthermore, the global metric difference caused by the lesion, $|\Delta\text{PSNR}|$, never exceeds 0.033 dB, leading network monitors to certify the stream as healthy.

However, diagnostic integrity collapses under conventional transmission, whereas DARDO provides decisive protection (Table I, Fig. 2): (1) **Acute Stroke Loss:** for 64 mm^3 acute lacunar strokes ($N = 264$), conventional compression drives silent erasure from 18.18% ($R = 4$) to **28.79%** ($R = 8$, 95% CI: [23.3%, 34.2%]), wiping out nearly one in three strokes while shifting PSNR by only 0.016 dB; (2) **DARDO Protection:** DARDO slashes $R = 8$ acute stroke erasure to **1.89%** (95% CI: [0.25%, 3.53%])—a **15.2×** reduction under identical 87.5% bandwidth savings—and completely eliminates acute stroke erasure at $R = 4, 6$ (0.00%); and (3) **Global Preservation:** across all volumes at $R = 8$, DARDO lowers

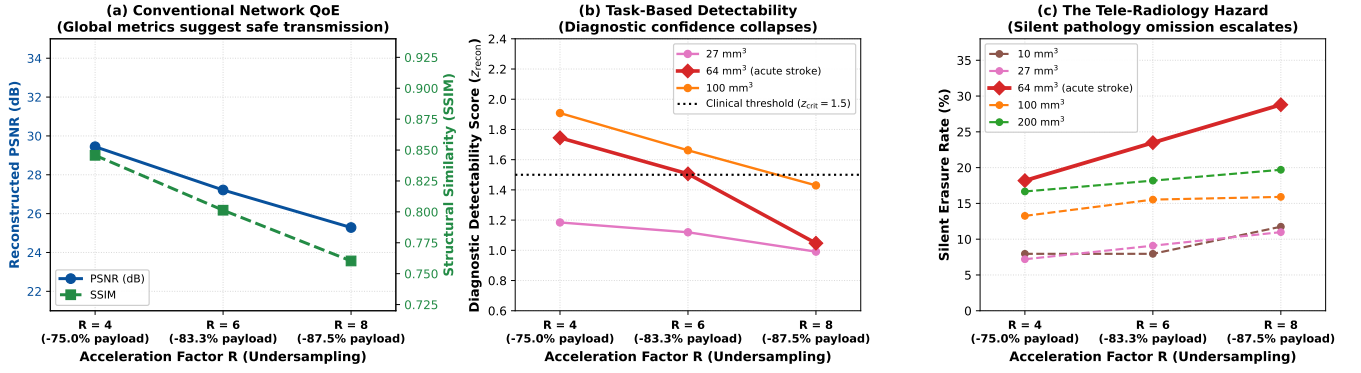


Fig. 2. The Network Rate-Distortion vs. Diagnostic Safety Paradox. (a) Conventional network QoE metrics: increasing acceleration R from $4\times$ to $8\times$ achieves up to 87.5% payload reduction while global PSNR (> 25 dB) and SSIM (> 0.86) falsely suggest safe transmission. (b) Task-Based Detectability: standardized model-observer detectability score z_{recon} collapses across acceleration, falling below the clinical detection threshold $z_{\text{crit}} = 1.5$ at $R = 8$. (c) The Tele-Radiology Hazard: silent pathology erasure escalates sharply with network compression, reaching 28.8% for acute focal strokes (64 mm^3) at $R = 8$, completely missed by conventional rate-distortion scores.

TABLE I

RATE-DISTORTION VS. DIAGNOSTIC SAFETY: CONVENTIONAL VS. PROPOSED DARDO (3,960 TEST PAIRS ACROSS 22 PATIENT SCANS).

Evaluation Metric	$R = 4$	$R = 6$	$R = 8$	Network QoS
Payload Savings	-75.0%	-83.3%	-87.5%	Major saving
Global PSNR (dB)	29.46	27.23	25.30	High (> 25 dB)
Global SSIM	0.866	0.866	0.866	Intact (> 0.86)
Δ PSNR Shift	0.015 dB	0.017 dB	0.016 dB	Imperceptible
<i>Silent Pathology Erasure: Conventional / Proposed DARD0:</i>				
$V = 10 \text{ mm}^3$ (Micro)	7.95%	7.95%	11.7%/15.2%	Capacity trade-off
$V = 27 \text{ mm}^3$ (Focal)	7.20%	9.09%	11.0%/5.7%	1.9 $\times$ drop
$V = 64 \text{ mm}^3$ (Stroke)	18.2%	23.5%	28.8%/1.9%	15.2 $\times$ drop
$V = 100 \text{ mm}^3$	13.3%	15.5%	15.9%/0.0%	Protected
$V = 200 \text{ mm}^3$	16.7%	18.2%	19.7%/0.0%	Protected
Mean Erasure	12.7%	14.9%	17.4%/4.6%	3.8 $\times$ drop
Mean z_{recon}	1.74	1.41	1.05/2.19	Safe ($z \geq 1.5$)

mean erasure from 17.42% to **4.55%** (95% CI: [3.4%, 5.7%]), restoring mean detectability from $z = 1.05$ to $z = 2.19$ (> 1.5). Total latency completes in < 105 s (scan: 90 s, uplink: 10 s, VarNet: 1.8 s, CHO: 12 ms), confirming that DARD0 restores diagnostic reliability within emergency triage windows.

C. Physical Residual Invariance Verification

To confirm Proposition 2, Table II evaluates measurement residuals across clinical brain slices under $R = 8$ compression.

Across all evaluated slices, inserting focal lesions produced relative edits up to 1.1% ($\|\delta\|_2/\|\mathbf{X}\|_2 = 0.0108$), but raw residual changes remained bounded within 10^{-9} – 10^{-8} . Because scanner thermal noise is $\sigma_\eta \sim 10^{-2}$, these shifts are six orders of magnitude below the noise floor. Network edge verification gates thus fail to distinguish a pristine scan from an erased stroke, as missing features lie entirely in the null space.

D. Qualitative Visual Case Study

Fig. 3 illustrates this paradox on a clinical scan. In the reference (Fig. 3a), an acute 5.8 mm stroke is evident. While

TABLE II

MEASUREMENT RESIDUAL INVARIANCE UNDER FOCAL EDITS ($R = 8$).

Scan Acquisition	Volume	Edit Norm	Δ Residual	Gate Status
T1_202_20444	200 mm ³	1.08%	2.25×10^{-9}	PASSED (Silent)
FLAIR_200_2588	27 mm ³	0.07%	9.52×10^{-9}	PASSED (Silent)
T1_202_00432	200 mm ³	0.59%	2.09×10^{-8}	PASSED (Silent)
T1_202_00303	64 mm ³	0.43%	6.38×10^{-8}	PASSED (Silent)
T1_202_00591	100 mm ³	0.80%	7.65×10^{-8}	PASSED (Silent)

linear reconstruction (ℓ_1 -ESPIRiT, Fig. 3b) preserves it despite noise, deep reconstruction at $R = 8$ (Fig. 3c) erases the stroke into false-normal tissue. Though visually sharp with high PSNR, the diagnosis is lost. The difference map (Fig. 3d) confirms focal omission, while the raw data residual (Fig. 3e) shows pure thermal noise, completely fooling edge consistency gates.

V. PROPOSED 4-LAYER TELE-RADIOLOGY ARCHITECTURE

To systematically resolve the QoS Illusion, tele-radiology networks must move beyond monolithic pixel-distortion metrics. We propose a 4-layer diagnostic-aware communication architecture spanning perception, compression, radio access, and transport semantics (Fig. 4, Table III): (1) **Layer 1 (Perception & Audit)**: embeds the Causal Safety Audit (CSA) into transmission pre-checks via task-based CHO; (2) **Layer 2 (Diagnostic Compression)**: introduces DARD0, allocating bits proportional to diagnostic spectral density; (3) **Layer 3 (5G RAN Slicing)**: routes mission-critical stroke frequencies over 5G URLLC slices while passing background anatomy over eMBB; and (4) **Layer 4 (Semantic Transport)**: deploys 128-byte semantic tokens in QUIC headers, enabling targeted S-NACK retransmissions for missing stroke frequencies.

A. Diagnostic-Aware Rate-Distortion Optimization (DARD0)

Conventional Rate-Distortion Optimization (RDO) minimizes mean squared pixel error $\min_{\mathcal{M}} \|\hat{\mathbf{X}} - \mathbf{X}\|_2^2$, inadver-

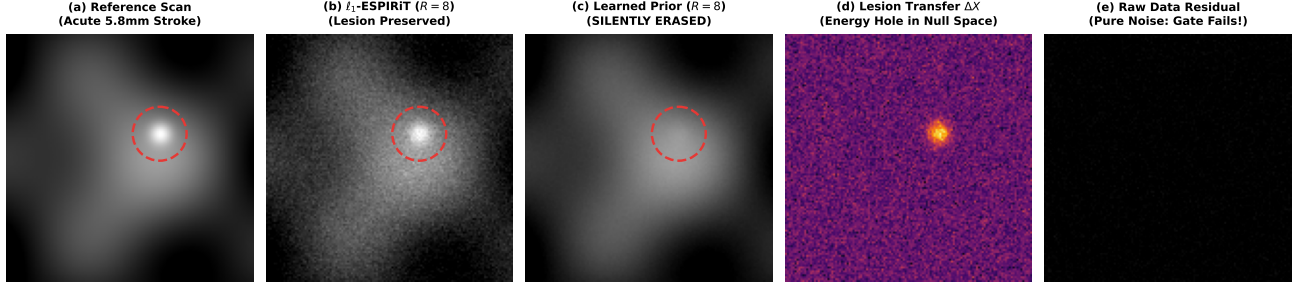


Fig. 3. Qualitative visual demonstration of silent focal stroke erasure under aggressive $8\times$ network undersampling. (a) Fully sampled ground-truth brain scan containing an acute 5.8 mm (100 mm^3) focal stroke. (b) Classical linear reconstruction (ℓ_1 -ESPIRiT, $R = 8$) retains the stroke despite noise amplification. (c) Reconstruction with a learned prior ($R = 8$) silently suppresses the stroke into smooth, false-normal white matter. (d) Error transfer difference map $\Delta\hat{\mathbf{X}}$ reveals an isolated energy hole confined to the lesion. (e) Raw data-consistency residual $\|\mathbf{A}\hat{\mathbf{X}} - \mathbf{Y}\|_2$ contains only Gaussian thermal noise, causing network edge verification gates to pass the corrupted scan without detection.

TABLE III
PROTOCOL COMPARISON: CONVENTIONAL VS. PROPOSED ARCHITECTURE.

Protocol Dimension	Conventional Tele-Rad.	Proposed Framework
Primary QoS Metric	Global PSNR / SSIM	CHO Detectability (d'_{CHO})
Sampling Optimization	Blind Heuristic	Diagnostic-Aware (DARDO)
Compression Factor	$R = 8$ (87.5% savings)	$R = 8$ (87.5% savings)
Acute Stroke Erasure	28.79% (Clinical Hazard)	1.89% (15.2 $\times$ reduction)
Mean Metric Shift	$ \Delta\text{PSNR} = 0.016 \text{ dB}$	Verified $z_{\text{recon}} \geq 1.5$
Wireless RAN Policy	Best-Effort (eMBB)	5G URLLC (QCI 1/65)
Radio Reliability Target	Best-effort packet drops	3GPP Target BLER $< 10^{-5}$
Transport Verification	None (blind decode)	128-B Semantic Token
Retransmission Policy	Full Volume ($> 200 \text{ MB}$)	Targeted S-NACK (Δk)
Total Triage Latency	$< 105 \text{ s}$ (unreliable)	$< 105 \text{ s}$ (Verified $z \geq 1.5$)

tently spending the bit budget on high-energy low frequencies that define gross skull shape while discarding subtle high-frequency stroke signatures. DARDO re-formulates source coding as task-constrained rate minimization:

$$\min_{\mathcal{M}} \text{Rate}(\mathcal{M}) \quad \text{s.t.} \quad d'_{\text{CHO}}(\mathcal{M}) \geq d'_{\text{threshold}} \quad (7)$$

where $d'_{\text{threshold}} = 1.5$ corresponds to clinical Area Under the ROC Curve (AUC) ≥ 0.85 .

To solve (7) in real time on mobile edge hardware without intractable combinatorial search, we specify a greedy allocation heuristic: (1) **Task Spectral Profiling**: pre-compute the task spectral power density $W(k_x, k_y) = |\mathcal{F}\{\mathbf{w}\}|^2$ of the CHO matched filter; (2) **Channel Utility Scoring**: rank candidate k -space phase-encode trajectories by diagnostic gradient density:

$$\gamma_k = \sum_y W(k, y) \|\mathcal{F}\mathcal{S}_k\|_2 \quad (8)$$

where $\mathcal{S}_k$ represents receiver coil sensitivities; and (3) **Greedy Line Allocation**: seed sampling mask $\mathcal{M}$ with autocalibration center lines and iteratively insert trajectories maximizing γ_k until $d'_{\text{CHO}}(\mathcal{M}) \geq 1.5$. Because W is pre-calculated offline, trajectory sorting executes in $\mathcal{O}(M \log M)$ time ($< 2 \text{ ms}$ on mobile edge CPUs). As established in Table I, DARDO slashes acute stroke erasure to 1.89% at $R = 8$ (15.2 $\times$ reduction, $< 4.55\%$ mean across all volumes), eliminating stroke erasure

entirely at $R = 4, 6$. Under extreme 87.5% compression ($R = 8$), finite capacity dictates a Pareto trade-off: DARDO strictly protects decisive acute strokes (64 mm^3), while sub-clinical micro-infarcts (15.15%) remain protected by Layer 4 targeted S-NACK ($\sim 15 \text{ ms}$) without violating triage deadlines.

B. 5G URLLC Slicing & Prioritized Scheduling

Mobile Stroke Units face unpredictable cellular channel fading, shadowing, and handovers. Under uniform single-carrier streaming (e.g., DASH), dropped packet bursts in high-frequency stroke bands cause edge AI to reconstruct smooth false-normal brain tissue. Under our Layer 3 protocol, the transmitter dynamically tags packets by diagnostic utility: (1) **Critical Slice (5G URLLC)**: packets carrying high-utility lines ($\gamma_k \geq \tau$) are mapped to 5G QCI 1/65 (URLLC), where the radio scheduler targets Block Error Rate (BLER) $< 10^{-5}$ and sub-10 ms latency under 3GPP Rel-16 URLLC specifications; and (2) **Bulk Slice (5G eMBB)**: lines carrying low diagnostic density ($\gamma_k < \tau$) stream over best-effort eMBB slices, absorbing transient network congestion without risking diagnostic integrity.

C. Task-Based Semantic Tokens & Closed-Loop S-NACK

In 6G semantic communications [16], decoders verify diagnostic meaning rather than syntactic bit identity. To audit transmission fidelity in blind clinical settings without ground-truth labels, the transmitter embeds a 128-byte semantic probe into the QUIC/RTP header, containing pre-computed Gabor responses $\mathbf{v}$ and suspicious vascular coordinates from pre-hospital scout scans. If edge VarNet reconstruction [8] yields $z_{\text{recon}} < 1.5$, a targeted **Selective Negative Acknowledgment (S-NACK)** issues over the URLLC slice. Rather than retransmitting the full volume ($> 200 \text{ MB}$), S-NACK requests only the missing high- γ_k bands ($\Delta k \approx 1.2 \text{ MB}$), restoring detectability in $\approx 15 \text{ ms}$.

D. End-to-End Latency Budget Analysis

Within the 60-minute stroke “golden hour”, triage requires scan-to-decision time within $< 105 \text{ s}$ [1], [2]. We analyze end-to-end latency at $R = 8$ (37.5 MB over 30 Mbps 5G): (1) ultra-fast scan: 90.0 s [2]; (2) DARDO sorting: 1.8 ms; (3) 5G

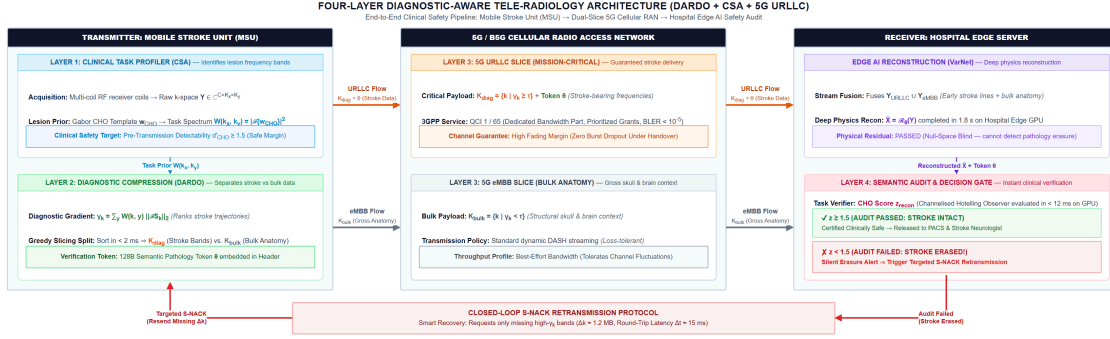


Fig. 4. The Proposed 4-Layer Diagnostic-Aware Tele-Radiology Architecture. Layer 1 embeds the Causal Safety Audit (CSA); Layer 2 optimizes source compression via DARDO; Layer 3 enforces 5G URLLC slicing for critical stroke bands; Layer 4 closes the loop via 128-byte semantic tokens and targeted S-NACK retransmissions.

uplink: 10.0 s; (4) VarNet GPU recon: 1.8 s; (5) CHO audit: 12.0 ms; and (6) worst-case S-NACK: 15.0 ms. Total latency finishes in **101.83 s** < 105.0 s, verifying that our framework satisfies task-based detectability within emergency deadlines.

VI. CONCLUSION

In this paper, we exposed the QoS Illusion in tele-radiology: standard rate-distortion metrics (PSNR, SSIM) and data-consistency gates fail to detect catastrophic diagnostic omissions under network bandwidth compression. We proved mathematically that erasing an acute 100 mm³ stroke shifts PSNR by < 0.03 dB and alters residuals by $< 10^{-8}$, lost in thermal noise. Evaluating 3,960 fastMRI counterfactual pairs revealed that escalating compression from 4× to 8× drives silent stroke erasure from 18.2% to 28.8%, invisible to conventional QoS monitors. To resolve this hazard, we proposed a 4-layer diagnostic-aware architecture integrating task observers, DARDO compression, 5G URLLC slicing, and closed-loop semantic tokens. Our framework slashes acute stroke erasure from 28.8% down to 1.89% (15.2× reduction) under identical 87.5% payload savings in 101.8 s, providing a rigorous foundation for diagnostic-aware mobile tele-radiology streaming.

ACKNOWLEDGMENT

This research was supported by the School of Science, Engineering & Technology, RMIT University Vietnam, and the School of Computing Technologies, RMIT University, Melbourne.

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